

# Reasoning Commitment Points in Transformer Chain-of-Thought

A Causal Mechanistic Study of Trajectory Lock-In

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## Abstract

Chain-of-thought (CoT) prompting enables large language models to escape the computational limits of a single forward pass, empirically reaching difficult reasoning problems and theoretically expanding expressivity toward polynomial-time computation. However, when a model begins on an incorrect reasoning path, it almost never self-corrects—it propagates and amplifies the error through subsequent steps. This work investigates the mechanistic cause of that lock-in. The central question is: *at what semantic stage of a CoT trace does an incorrect trajectory become causally unrecoverable, even under direct residual-stream intervention?* This is empirically addressed using activation patching—forcibly replacing the internal hidden states of a wrong-CoT run with those from a correct-CoT run at specific semantic stages (Setup, Computation, Transition, Conclusion), then measuring whether the final answer flips. Pilot results on  $N = 2$  pairs suggest commitment is front-loaded: setup and computation patching shifts logit mass toward the correct answer by +14 points; transition patching yields only +0.3. If this pattern replicates at  $N = 80$ , it implies that the effective correction window closes before the model writes logical-connector phrases such as “therefore,” with direct implications for where steering, verification, and monitoring tools can actually operate.

**Keywords:** mechanistic interpretability • activation patching • chain-of-thought • reasoning commitment • GSM8K • Qwen2.5-14B

*This document includes the technical report (Sections 1–8) and an annotated reading list for collaborators (Appendix A).*

## 1 Introduction

Large language model (LLM) reasoning has made substantial empirical progress through chain-of-thought (CoT) prompting, in which a model generates intermediate reasoning steps before producing a final answer [Wei et al., 2022]. On multi-step arithmetic benchmarks such as GSM8K, CoT dramatically improves accuracy over direct-answer prompting, especially at scale.

Theoretically, this is well-motivated: a single forward pass of a constant-depth transformer is bounded by

circuit complexity class  $TC^0$ , but generating  $T$  intermediate tokens provides  $T$  additional computational steps, with linear CoT yielding recognizable context-sensitive languages and polynomial CoT capturing exactly P [Merrill & Sabharwal, 2024; Feng et al., 2023].

Despite this expanded computational power, CoT exhibits a striking failure mode: once a model takes an incorrect reasoning step, it almost never recovers. It continues generating, building on its error, arriving at a wrong answer with apparent confidence. This persistence of incorrect trajectories is not explained by existing complexity theory, which only establishes that the total computation across a trace is *sufficient* for correct reasoning—it says nothing about where in the trace the critical computation sits, or whether the trajectory is reversible after a mistake.

This project fills that gap empirically. The question is reframed as a causal intervention problem: at which semantic stage does an incorrect CoT trajectory become unrecoverable, meaning that even forcibly replacing the model’s internal state with the correct-run state at that stage fails to rescue the answer?

## 2 Background

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### 2.1 Chain-of-Thought Prompting and GSM8K

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CoT prompting was introduced by Wei et al. (2022) as a few-shot method that substantially improves LLM performance on arithmetic, commonsense, and symbolic reasoning tasks, particularly for very large models. The GSM8K benchmark [Cobbe et al., 2021] is a collection of 8,500 grade-school math word problems with CoT-style annotated solutions; it has become the standard evaluation environment for multi-step arithmetic reasoning.

### 2.2 Complexity-Theoretic Motivation

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A standard constant-depth decoder-only transformer without CoT is bounded by  $TC^0$ —a shallow circuit complexity class that supports pattern matching and factual recall but cannot solve problems requiring sequential or iterative computation [Merrill & Sabharwal, 2024]. Chain-of-thought changes the computational picture:

- With  $O(n)$  steps, the model escapes  $TC^0$  and can recognize all regular languages.
- With polynomial steps, expressivity exactly matches P [Merrill & Sabharwal, 2024].
- Bounded-depth transformers cannot solve basic arithmetic directly; constant-size autoregressive transformers with CoT derivations can, and can also solve classes of dynamic programming problems [Feng et al., 2023].

These results establish that a CoT trace carries sufficient computational budget to solve hard reasoning problems. They do not characterize where in the trace this budget is exercised, nor whether a wrong prefix locks the trajectory before the budget is spent.

### 2.3 Activation Patching as a Causal Tool

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Activation patching—also known as causal tracing or interchange intervention—is a mechanistic interpretability technique for establishing causal responsibility of model components [Meng et al., 2022]. In the canonical formulation:

1. Run a “clean” (correct) input through the model; save all internal hidden states.

2. Run a “corrupted” (incorrect) input; at a component of interest, replace its state with the clean state.
3. Measure whether the output changes toward the clean output.

A component that, when restored, rescues the output is causally responsible for that output difference. Best practices—including metric choices and corruption design—are documented in Zhang & Nanda (2024) and Heimersheim & Nanda (2024).

### 3 Research Question

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The central question is stated precisely:

*At which semantic stage of a CoT trace does an incorrect reasoning trajectory become unrecoverable, meaning that even forcibly replacing the model’s internal state with the correct-run state at that stage fails to rescue the answer?*

“Unrecoverable” is defined operationally: patching the full residual stream of the wrong-CoT run with states from a correct-CoT run, at all tokens in a given stage and all layers simultaneously, does not cause the model to flip its final answer from wrong to correct. This is a causal question, not an observational one. A stage that, when patched, rescues answers contains causally necessary information for the correct trajectory.

**Running example:** A model solving “Janet earns \$20/day and spends \$5/day. How much does she save in a week?” (correct: \$105) makes an arithmetic error computing  $20 - 5 = 10$  instead of 15, arriving at \$70. If patching the Setup stage rescues the answer, commitment had not yet occurred at Setup. If patching the Transition stage does nothing, commitment had already occurred before the model wrote “therefore.”

### 4 Method

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#### 4.1 Model and Hardware

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The model used is **Qwen2.5-14B-Instruct**, quantized to NF4 (4-bit) and run on a single NVIDIA T4 GPU (Kaggle free tier). At 14B parameters, the model is capable of complex multi-step reasoning while remaining feasible on single-GPU hardware. NF4 quantization reduces memory footprint with minimal impact on reasoning accuracy at this scale.

#### 4.2 Dataset: GSM8K Hard Tail

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The evaluation set is problems 900–1319 from the GSM8K test split (the “hard tail”)—419 problems on which the model fails most frequently under greedy decoding. This ensures a sufficient fraction of model runs produces incorrect CoT traces.

#### 4.3 Constructing Paired Traces

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**Correct trace:** Generated by greedy decoding ( $T = 0$ ). On hard-tail problems where the model solves deterministically, the greedy output constitutes a clean, noise-free correct reasoning chain.

**Wrong trace:** Generated by temperature sampling ( $T = 1.0$ – $1.4$ ), repeating until an incorrect answer is produced (verified against GSM8K ground truth). Sampled wrong traces follow a genuinely incorrect reasoning path throughout.

This is critical: greedy failures are structurally unsuitable for this experiment. On hard problems, greedy failures tend to be soft failures—the model reasons correctly but outputs a wrong final token against the grain of its own residual stream. Teacher-forced forward passes on such traces already predict the correct answer before any patching, making them useless for commitment analysis. Sampled wrong traces do not have this property.

This yields matched pairs: same problem, same model, same prompt—differing only in which reasoning path the model took. The full run targets  $N = 80$  organic pairs.

#### 4.4 Stage Classification

The incorrect trace is segmented into four semantic stages based on surface text patterns:

| Stage              | Definition                                     | Examples                                 |
|--------------------|------------------------------------------------|------------------------------------------|
| <b>Setup</b>       | Restatement of problem variables and structure | “Janet earns \$20 a day...”              |
| <b>Computation</b> | Arithmetic operations, numeric manipulations   | “ $20 - 5 = \dots$ ”, numbers, operators |
| <b>Transition</b>  | Logical connectors between steps               | “Therefore”, “So”, “Thus”, “Hence”       |
| <b>Conclusion</b>  | Tokens after final transition through answer   | “The answer is ##### 105”                |

Table 1. Stage definitions. Tokens not matching Computation, Transition, or Conclusion are assigned to Setup.

#### 4.5 Activation Patching Procedure

For each pair and each stage  $S$ :

1. Run the correct trace through the model; cache residual-stream activations at all 48 layers for all token positions.
2. Run the wrong trace. At every token position belonging to stage  $S$ , replace the residual-stream state at all layers with the cached correct-trace state (using proportional position mapping to align sequences).
3. Resume the forward pass and collect the model’s output.

Pairs where the baseline teacher-forced forward pass on the wrong trace already assigns higher logit probability to the correct answer are excluded—these exhibit soft failures and are structurally unsuitable for commitment measurement.

#### 4.6 Metrics

**Flip rate:** Fraction of pairs in which patching stage  $S$  causes the final answer to change from wrong to any other value. A high flip rate at stage  $S$  indicates  $S$  carries causally recoverable information.

**Logit diff shift ( $\Delta_{ld}$ ):** Change in log-probability mass attributed to the correct answer after patching, versus the unpatched wrong run. A continuous metric that captures partial recovery even when no full flip occurs.

The commitment point is the earliest stage after which both metrics collapse to near zero.

## 5 Pilot Results

***Caution:** All results are from  $N = 2$  pairs. Numbers are directional only and must not be interpreted as estimates of true effect sizes. Statistical analysis requires  $N = 80$ .*

The pilot measurement of primary interest is  $\Delta_{ld}$ —the change in log-probability mass toward the correct answer after patching:

| Stage       | $\Delta_{ld}$ (logit diff shift) | Signal     |
|-------------|----------------------------------|------------|
| Setup       | +14.6                            | Strong     |
| Computation | +13.9                            | Strong     |
| Transition  | +0.3                             | Negligible |
| Conclusion  | —                                | No data    |

Table 2. Logit difference shift per semantic stage ( $N = 2$  pilot, directional only). Flip rates not reported at  $N = 2$  ( $\leq 1$ –2 observations per stage).

Setup and computation patching shift logit mass strongly toward the correct answer. Transition patching is near-zero. Conclusion has no data at this sample size.

## 6 Discussion

### 6.1 Commitment Is Front-Loaded

The pilot results suggest that the information determining the final answer is concentrated in the Setup and Computation stages. Logical-connector phrases (Transition stage) follow a commitment already encoded in the residual stream. This is counterintuitive: from the outside, “therefore” looks like a reasoning step. Internally, it may be better characterized as an announcement of a conclusion already reached—pending validation at  $N = 80$ .

The pilot data show that by the transition stage, patching all 48 layers simultaneously no longer shifts the answer toward correct—even under maximum intervention. Whether this reflects genuine irrecoverable commitment or an error distributed across layers in a way that simultaneous patching cannot access is exactly what the per-layer analysis (Week 3) will test.

This connects to Dutta et al. (2024), who identified a “functional rift” in middle layers of CoT-using LLMs—early layers more influenced by pretraining priors, later layers by in-context CoT content. Similarly, Mehrafarin et al. (2026) show that task-relevant information appears earlier in GSM8K CoT traces and concentrates in mid-to-late layers—consistent with commitment preceding visible inference-linking steps.

## 6.2 Implications for Correction and Monitoring

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If commitment is established at Setup/Computation stages, any correction mechanism must act before the model writes arithmetic expressions. Monitoring or steering tools that intervene at transition tokens are operating too late. This directly constrains:

- **Process reward models** [Lightman et al., 2023]: step-level rewards at transition or conclusion tokens may be unable to alter already-committed trajectories.
- **Activation steering**: interventions must target early-to-mid generation positions, not logical-conclusion text.
- **Real-time monitoring**: a practical system must evaluate setup quality rather than reasoning-chain internal consistency.

## 6.3 Limitations

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**Surface-text proxy.** Stage labels are derived from surface text patterns. Hard computation may occur internally while “therefore” is being generated—the stage labeling is a proxy, not ground truth. The per-layer analysis (§7) will test this: if computation signal concentrates in layers 5–20 and transition signal in layers 35–48 without overlap, the proxy is tracking something real.

**Stage aggregation.** The four-stage segmentation collapses the cyclic structure of multi-step CoT into flat buckets. Whether commitment happens at the first wrong step or accumulates across rounds requires occurrence-indexed patching ( $\text{Computation}_1, \text{Computation}_2, \dots$ ), left to future work.

**Pair construction.** The correct trace is greedy; the wrong trace is temperature-sampled. Proportional position mapping is an approximation. Sensitivity to this design choice is left to future work.

**Pilot sample size.**  $N = 2$  is insufficient for statistical conclusions. All pilot numbers are directional only. The full  $N = 80$  run with 95% bootstrap confidence intervals will determine robustness.

**Single model.** Results are from Qwen2.5-14B-Instruct (NF4). Generalization across model families, scales, and training regimes is an open question.

**Cross-problem null control.** The current design patches correct-trace residual states from the *same* problem into the wrong run. There is no null control using residual states from a *different* problem’s correct trace. Without this baseline, the observed  $\Delta_{\text{ld}}$  recovery could partially reflect generic “correct-reasoning-trajectory” signal rather than problem-specific committed content. A cross-problem null would isolate the problem-specific component of the commitment signal.

## 7 Next Steps

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| Week                    | Task                                                                                                                                                                                                      |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Week 1</b> (current) | Full pair collection: 80 greedy-correct traces + 80 temperature-sampled wrong traces from GSM8K problems 900–1319.                                                                                        |
| <b>Week 2</b>           | Full patching sweep; compute flip rate and $\Delta_{\text{Id}}$ curves with 95% bootstrap CIs over $N = 80$ pairs.                                                                                        |
| <b>Week 3</b>           | Per-layer patching on the highest-signal stage (expected: Setup or Computation). Patch one layer at a time. Prediction: commitment signal in layers 5–20; post-commitment in layers 35–48 for Transition. |
| <b>Week 4</b>           | 2-page technical report + arXiv submission.                                                                                                                                                               |

## 8 Related Work

**CoT expressiveness theory.** Merrill & Sabharwal (2024) and Feng et al. (2023) establish that CoT lifts constant-depth transformers from  $\text{TC}^0$  to polynomial-time expressivity. These are existence results and do not address trajectory commitment.

**CoT faithfulness.** Lanham et al. (2023) and Turpin et al. (2023) motivate the shift to hidden-state interventions: CoT text faithfulness is unreliable, but residual states are the ground truth. Chen, Benton et al. (2025, Anthropic) extend this to frontier reasoning models.

**Mechanistic CoT analysis.** Dutta et al. (2024) find parallel answer-generation pathways and a mid-layer functional rift. Cabannes et al. (2024) identify “iteration heads” implementing step-to-step information carrying. Both characterize which circuit components participate in CoT but do not study commitment points.

**Arithmetic mechanistic analysis.** Stolfo, Belinkov & Sachan (2023) apply causal mediation analysis directly to arithmetic reasoning in transformers, finding task-relevant information flows from mid-sequence early layers to the final token via attention. This is the direct methodological predecessor on single-step arithmetic; our work extends the intervention to multi-step CoT and introduces the commitment-point framing.

**Hidden-state patching on GSM8K CoT.** Mehrafarin et al. (2026) perform the closest experiment: token-level activation patching on GSM8K. Key differences: (1) CoT→direct-answer, not wrong-CoT→correct-CoT; (2) individual tokens, not semantic stages; (3) no commitment point defined or measured.

**Probe-based commitment evidence.** Afzal et al. (2025) show that a linear classifier on LLM hidden states predicts CoT success before the first reasoning token. Probing (observational) and patching (causal) pointing to the same conclusion from different methodological angles strengthens the commitment-point claim.

**Sentence-level importance.** Bogdan, Macar, Nanda & Conmy (2025, Thought Anchors) find that planning and uncertainty-management sentences have disproportionate causal impact on final answers, while arithmetic sentences often do not—in apparent tension with the pilot result that Computation patching yields high  $\Delta_{\text{Id}}$ . A direct comparison is warranted.

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*This is a work-in-progress technical report. Full results and statistical analysis will appear in the final version.*

## Reading List

### Reasoning Commitment Points in Transformer Chain-of-Thought

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Prepared by Aditya Singh—IIT Kharagpur / University of Birmingham. For collaborators who want to understand the project context. A reader with ML fundamentals can build full working understanding by reading in order. Skip to §4–5 if you already know ML basics.

#### Section 1. Background

*The empirical and dataset papers that set up the problem.*

##### 1.1 Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

*Wei et al. — NeurIPS 2022 — arXiv:2201.11903*

**Why:** The paper that introduced CoT prompting and showed it dramatically improves LLMs on GSM8K, MultiArith, and commonsense benchmarks. Read this first.

##### 1.2 Training Verifiers to Solve Math Word Problems (GSM8K)

*Cobbe et al. — OpenAI 2021 — arXiv:2110.14168*

**Why:** Introduces the GSM8K benchmark—8.5k grade-school math word problems with CoT rationales. Our dataset. Explains the “hard tail” structure and why models fail on harder problems.

##### 1.3 Self-Consistency Improves Chain of Thought Reasoning

*Wang et al. — ICLR 2023 — arXiv:2203.11171*

**Why:** Introduces sampling multiple CoT paths and picking the most consistent answer. Relevant because temperature sampling to generate wrong traces follows the same stochastic multi-path idea—the model takes different reasoning routes from the same prompt.

#### Section 2. Theoretical Foundations

*The formal backbone for the “single pass =  $TC^0$ ,  $CoT \rightarrow P$ ” story.*

##### 2.1 The Expressive Power of Transformers with Chain of Thought

*Merrill & Sabharwal — ICLR 2024 — arXiv:2310.07923*

**Why:** Formally proves that constant-depth transformers are bounded ( $TC^0$ ), but with polynomial CoT steps exactly capture P. Main citation for the  $TC^0 \rightarrow P$  claim.

## 2.2 Towards Revealing the Mystery behind Chain of Thought: A Theoretical Perspective

Feng et al. — NeurIPS 2023 — arXiv:2305.15408

**Why:** Independent complexity argument: bounded-depth transformers cannot solve basic arithmetic directly, but autoregressive transformers with CoT can. Also links CoT to dynamic programming. Complements Merrill & Sabharwal.

## 2.3 Chain of Thought Empowers Transformers to Solve Inherently Serial Problems

Li et al. — ICLR 2024 — arXiv:2402.12875

**Why:** Shows CoT specifically helps with inherently serial tasks provably beyond constant-depth transformers. Good empirical grounding for the theory.

# Section 3. Mechanistic Interpretability Foundations

*Methodological backbone. Understand activation patching rigorously before reading Sections 4–5.*

## 3.1 A Mathematical Framework for Transformer Circuits

Elhage et al. — Anthropic / Transformer Circuits Thread 2021

**Why:** Foundational MI vocabulary: residual stream, attention heads, MLPs, superposition, the “residual stream as communication bus” view. Required background before any patching experiment.

## 3.2 Locating and Editing Factual Associations in GPT (ROME)

Meng et al. — NeurIPS 2022 — arXiv:2202.05262

**Why:** Introduced activation patching to locate factual knowledge. Original corrupt-one-run, restore-clean-state, measure-restoration design. Our patching procedure follows this directly.

## 3.3 Towards Best Practices of Activation Patching in Language Models

Zhang & Nanda — ICLR 2024 — arXiv:2309.16042

**Why:** The “how not to fool yourself” paper. Shows that corruption method, metric choice, and patching scope all change conclusions. Directly justifies our choice of flip rate +  $\Delta_{\text{Id}}$ .

## 3.4 How to Use and Interpret Activation Patching

Heimersheim & Nanda — arXiv 2024 — arXiv:2404.15255

**Why:** Survey of patching variants. Key point: patching shows a component *can* influence behavior, not that it normally does—a key limitation to acknowledge in the discussion.

## 3.5 Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 small

Wang et al. — arXiv 2022 — arXiv:2211.00593

**Why:** Canonical circuit-level analysis using activation and path patching. Template for going from exploratory patching to rigorous circuit claims—the direction our per-layer analysis aims at.

# Section 4. Mechanistic Interpretability of CoT Specifically

*Papers studying CoT from the inside. All directly adjacent to this project.*

#### 4.1 How to Think Step-by-Step: A Mechanistic Understanding of Chain-of-Thought Reasoning

Dutta et al. — TMLR 2024 — arXiv:2402.18312

**Why:** First mechanistic study of CoT. Finds parallel answer-generation pathways and a “functional rift” in middle layers. Directly motivates the prediction that setup/computation and transition patches load onto different layer ranges.

#### 4.2 Iteration Head: A Mechanistic Study of Chain-of-Thought

Cabannes et al. — NeurIPS 2024 — arXiv:2406.02128

**Why:** Identifies “iteration heads” implementing step-to-step information carrying in CoT. Shows CoT capability can be traced to specific, transferable circuit components.

#### 4.3 A Mechanistic Interpretation of Arithmetic Reasoning in Language Models using Causal Mediation Analysis

Stolfo, Belinkov & Sachan — EMNLP 2023 — arXiv:2305.15054

**Why:** Direct methodological predecessor. Applies activation patching to arithmetic reasoning, finding task-relevant information flows from mid-sequence early layers to the final token. Same technique, same domain, single operations rather than multi-step CoT. Essential for positioning the contribution.

#### 4.4 How Does Chain of Thought Think? Mechanistic Interpretability with Sparse Autoencoding

Chen, Plaat & van Stein — arXiv 2025 — arXiv:2507.22928

**Why:** Feature-level patching on GSM8K using sparse autoencoders on Pythia-70M/2.8B. Note: scale is  $5\text{--}50\times$  smaller than Qwen2.5-14B; findings may not transfer directly.

#### 4.5 When Chain-of-Thought Fails, the Solution Hides in the Hidden States

Mehrafarin et al. — arXiv 2026 — arXiv:2604.23351

**Why:** Closest paper to our experiment. Token-level patching CoT→direct-answer on GSM8K. Key differences: (1) CoT→direct-answer, not wrong→correct CoT; (2) individual tokens, not semantic stages; (3) no commitment point defined or measured.

#### 4.6 Knowing Before Saying: LLM Representations Encode Information About CoT Success Before Completion

Afzal et al. — ACL Findings 2025 — arXiv:2505.24362

**Why:** Most direct external corroboration of the commitment-point hypothesis. Linear probes on hidden states predict CoT success before the first reasoning token. Probing (observational) and patching (causal) pointing to the same conclusion from different methodological angles strengthens the claim.

#### 4.7 Thought Anchors: Which LLM Reasoning Steps Matter?

Bogdan, Macar, Nanda & Conmy — arXiv 2025 — arXiv:2506.19143

**Why:** Counterfactual sentence-level importance for reasoning models: replace a sentence, resample, measure impact. Finds planning sentences have disproportionate causal impact; arithmetic sentences often do not—in apparent tension with the high Computation  $\Delta_{\text{Id}}$  in the pilot. A direct comparison is warranted.

## Section 5. Faithfulness, Supervision, and Steering

*Adjacent question: is the visible CoT causally connected to the answer, and can you steer it?*

## 5.1 Measuring Faithfulness in Chain-of-Thought Reasoning

Lanham et al. — Anthropic 2023 — arXiv:2307.13702

**Why:** Perturbs CoT text and measures answer changes. Larger models produce less faithful CoTs. Motivates using hidden-state patching rather than text manipulation.

## 5.2 Language Models Don't Always Say What They Think: Unfaithful Explanations in CoT

Turpin et al. — NeurIPS 2023 — arXiv:2305.04388

**Why:** CoT explanations can be sycophantic rationalizations diverging from the actual decision mechanism. Sets up the argument that “therefore” is a surface announcement of an already-made decision.

## 5.3 Reasoning Models Don't Always Say What They Think

Chen, Benton et al. (Anthropic Alignment Science) — arXiv 2025 — arXiv:2505.05410

**Why:** Extends Turpin to frontier reasoning models. Finds modern reasoning models arrive at correct answers through paths diverging from the visible CoT. Directly motivates why commitment-point monitoring must happen at the hidden-state level.

## 5.4 Let's Verify Step by Step (Process Reward Models)

Lightman et al. — OpenAI / ICLR 2024 — arXiv:2305.20050

**Why:** Trains process reward models to score individual CoT steps. Our results add a constraint: such verifiers must act on early stages—step-level rewards at transition/conclusion may be too late.

## 5.5 Chain-of-Thought Reasoning in the Wild Is Not Always Faithful

Arcuschin, Nanda et al. (Google DeepMind) — arXiv 2025 — arXiv:2503.08679

**Why:** CoT faithfulness is fragile in naturalistic settings. Contextualizes our work: commitment points matter for safety precisely because CoTs are not reliable representations of internal reasoning.

## Section 6. What Our Project Contributes

### Existing work covers:

- ✓ Theory: why CoT adds computational power (§2)
- ✓ Text-level faithfulness studies (§5.1, 5.2)
- ✓ Token-level patching on GSM8K (§4.4, 4.5)
- ✓ Layer-wise circuit characterization (§4.1, 4.2)
- ✓ Sentence-level black-box importance (§4.7)
- ✓ Probe-based early commitment evidence (§4.6)

### No existing paper:

- ★ Defines semantic stages from CoT surface text
- ★ Uses paired wrong/correct traces as intervention data
- ★ Full-residual-stream patching per stage (all 48 layers)
- ★ Measures flip rate &  $\Delta_{\text{ld}}$  vs. stage to locate a commitment point

**Our contribution: The first empirical localization of a stage-wise CoT commitment point via causal hidden-state intervention.**

## Quick Reference Table

| Paper                        | Sec. | arXiv                    | Purpose                             |
|------------------------------|------|--------------------------|-------------------------------------|
| Wei et al. 2022              | §1.1 | 2201.11903               | CoT introduced, GSM8K results       |
| Cobbe et al. 2021            | §1.2 | 2110.14168               | GSM8K dataset                       |
| Wang et al. 2023             | §1.3 | 2203.11171               | Self-consistency, CoT sampling      |
| Merrill & Sabharwal 2024     | §2.1 | 2310.07923               | $TC^0 \rightarrow P$ expressiveness |
| Feng et al. 2023             | §2.2 | 2305.15408               | $TC^0/DP$ theory                    |
| Li et al. 2024               | §2.3 | 2402.12875               | Serial tasks + CoT                  |
| Elhage et al. 2021           | §3.1 | transformer-circuits.pub | MI vocabulary                       |
| Meng et al. 2022 (ROME)      | §3.2 | 2202.05262               | Activation patching origin          |
| Zhang & Nanda 2024           | §3.3 | 2309.16042               | Patching best practices             |
| Heimersheim & Nanda 2024     | §3.4 | 2404.15255               | Patching interpretation             |
| Wang et al. 2022 (IOI)       | §3.5 | 2211.00593               | Circuit analysis template           |
| Dutta et al. 2024            | §4.1 | 2402.18312               | Layer roles in CoT                  |
| Cabannes et al. 2024         | §4.2 | 2406.02128               | Iteration heads circuit             |
| Stolfo et al. 2023           | §4.3 | 2305.15054               | Causal mediation on arithmetic      |
| Chen et al. 2025             | §4.4 | 2507.22928               | Feature patching on GSM8K           |
| Mehrafarin et al. 2026       | §4.5 | 2604.23351               | Token patching GSM8K (closest)      |
| Afzal et al. 2025            | §4.6 | 2505.24362               | Probing predicts CoT success        |
| Bogdan et al. 2025           | §4.7 | 2506.19143               | Thought Anchors                     |
| Lanham et al. 2023           | §5.1 | 2307.13702               | CoT faithfulness (Anthropic)        |
| Turpin et al. 2023           | §5.2 | 2305.04388               | Unfaithful CoT                      |
| Chen, Benton et al. 2025     | §5.3 | 2505.05410               | Faithfulness in frontier models     |
| Lightman et al. 2023         | §5.4 | 2305.20050               | Process reward models               |
| Arcuschin, Nanda et al. 2025 | §5.5 | 2503.08679               | CoT faithfulness at scale           |

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